

Satellite-Powered Flood Intelligence for Disaster Response

IIT Mandi Multimodal AI Hackathon 2026 · Track B — Mapping Flood Damage from Space

Summary

- End-to-end prototype: Sentinel-1 scene selection → 6-channel U-Net flood segmentation → historical-OSM infrastructure overlay → simplified road connectivity → dashboard and one-page situation report.
- Model (Kuro Siwo, event-separated): held-out test flood IoU 0.307, F1 0.470, precision 0.852, recall 0.324 on 94 samples from 6 unseen events.
- Trishuli, August 2026 (unvalidated inference): predicted flood extent 0.120 km² (0.55 % of analysed AOI pixels); 3 road ways and 36 buildings potentially affected, 2 bridges intersecting predicted flood, 2 of 3 settlements potentially cut off in a simplified road graph.

1 Problem and motivation

During a flood, responders need to know where water is, which roads, bridges and buildings may be affected, and which communities may have lost road access. In Himalayan valleys such as the Trishuli, floods coincide with monsoon cloud, so optical imagery is often unusable for weeks.

Sentinel-1 synthetic aperture radar (SAR) images through cloud, day and night, at about 10 m, and calm open water appears dark in its backscatter. Comparing a post-event image with pre-event images from the same orbit geometry highlights new water. VantageQ links that signal to pre-event OpenStreetMap infrastructure and a simple connectivity analysis, and presents results as indicators, not confirmed damage.

2 System overview

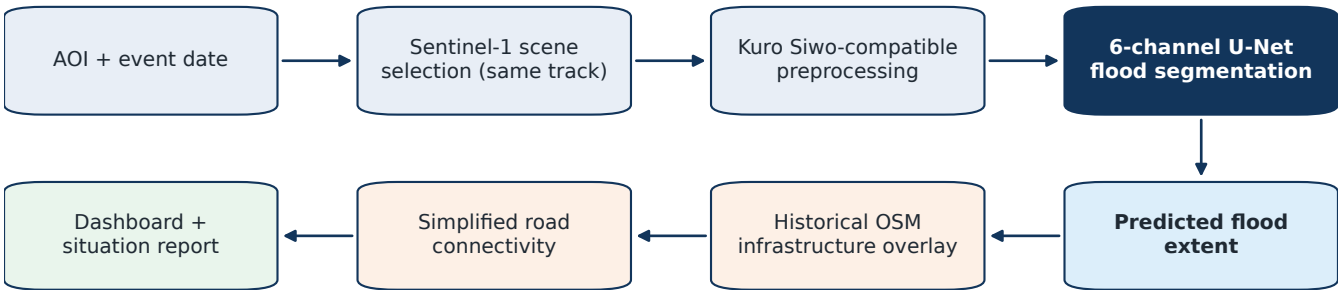


Figure 1. VantageQ workflow. Inputs: Sentinel-1 GRD via the Copernicus Data Space Ecosystem (orthorectified with Copernicus DEM inside Sentinel Hub), a U-Net trained on Kuro Siwo, and a pre-event OSM snapshot. Outputs are files read by the static dashboard and the report generator.

3 Data and compliance

Source	Role	Detail
Sentinel-1 GRD (IW, VV+VH)	Model input	post 2026-08-28, pre1 2026-08-16, pre2 2026-08-04; relative orbit 85 ascending
Kuro Siwo (labelled GRD)	Training/evaluation	252 train / 65 val / 94 test samples; official event splits
OpenStreetMap (historical)	Infrastructure	Overpass attic snapshot 2026-08-25T00:00:00Z (pre-event)
Copernicus DEM GLO-30	Orthorectification	used inside Sentinel Hub (COPERNICUS_30); not distributed

No Copernicus EMS product (including EMSR927), UNOSAT or other published damage/reference map, and no post-event OSM edit, was used as a production input, for training, thresholds or model selection. EMSR927 is reserved for a separate, not-yet-built evaluation-only comparison.

Attribution (docs/DATA_SOURCES.md §7):

Sentinel-1 / Sentinel-2 (Copernicus free & open data): Contains modified Copernicus Sentinel data 2026, processed by the VantageQ team

Copernicus DEM GLO-30 (Copernicus DEM GLO-30 licence (ESA), Article 6(b)): produced using Copernicus WorldDEM-30 © DLR e.V. 2010-2014 and © Airbus Defence and Space GmbH 2014-2018 provided under COPERNICUS by the European Union and ESA; all rights reserved; dataset citation: <https://doi.org/10.5270/ESA-c5d3d65>

Kuro Siwo dataset (CC BY 4.0): Cite Bountos et al., NeurIPS 2024

OpenStreetMap / ohsome (ODbL 1.0): © OpenStreetMap contributors

4 AI model and Kuro Siwo evaluation

- U-Net with a ResNet-18 encoder (ImageNet initialisation), the official Kuro Siwo baseline configuration.
- Input: 6 channels [post VV, post VH, pre1 VV, pre1 VH, pre2 VV, pre2 VH], linear σ^0 clamped to [0, 0.15] and normalised with fixed Kuro Siwo statistics (VV ($x - 0.0953$)/0.0427, VH ($x - 0.0264$)/0.0215); one shared preprocessing function for training and Trishuli inference.
- Output: 3 classes (no water, permanent water, flood); label 3 is excluded from loss and metrics (ignore_index=3).
- Training: 252 samples from 14 official train events; Adam (lr 1e-3), cosine schedule, flips, 20 epochs; checkpoint chosen on validation loss (65 samples, 4 validation events, best epoch 17). The checkpoint was then frozen.

Kuro Siwo split	Samples / events	Flood IoU	Flood F1	Precision	Recall
Validation (model selection)	65 / 4	0.560	0.718	0.657	0.791
Held-out test (evaluated once)	94 / 6	0.307	0.470	0.852	0.324

Held-out test, flood by event (sorted by IoU):

Event	Region	Samples	Flood IoU	Precision	Recall	Flood GT → perm. water
445	Romania	15	0.691	0.750	0.898	2.2 %
277	Greece	16	0.673	0.760	0.855	0.3 %
1111007	Nepal (lowlands, "Patna")	16	0.603	0.920	0.636	0.9 %
321	Honduras	16	0.192	0.664	0.213	4.1 %
1111013	USA (St Louis)	16	0.168	0.883	0.172	1.4 %
562	Australia (Burketown)	15	0.066	0.923	0.067	69.7 %

- These are Kuro Siwo dataset results on unseen events, not Trishuli accuracy. Kuro Siwo contains no steep Himalayan terrain.
- Flood precision is high but recall is low. 39.6 % of test flood pixels were predicted as permanent water, mostly in event 562, Australia (Burketown), which holds 56 % of test flood pixels.
- Validation scores are optimistic (same set used for selection). Not comparable with the Kuro Siwo paper (flood F1 80.1 % with full data, ResNet-50).

5 Trishuli case study (unvalidated inference)

- Event: Trishuli Valley flood, 26 August 2026; sub-AOI 85.17–85.22° E, 27.95–27.99° N near Betrawati.
- Sentinel-1 post 2026-08-28, pre1 2026-08-16, pre2 2026-08-04, all relative orbit 85 ascending (same viewing geometry).
- All dates on one shared EPSG:3857 grid (557 × 505 pixels of 10 map units), linear σ^0 (SIGMA0_ELLIPSOID), Lee 7×7, as in Kuro Siwo; tiled inference (224 px, overlap averaging; tiling difference 0.0029).
- Result: predicted flood extent 0.120 km², 0.55 % of the analysed AOI pixels. This is an unvalidated model prediction, not a measured flood area; no Trishuli ground truth was used.

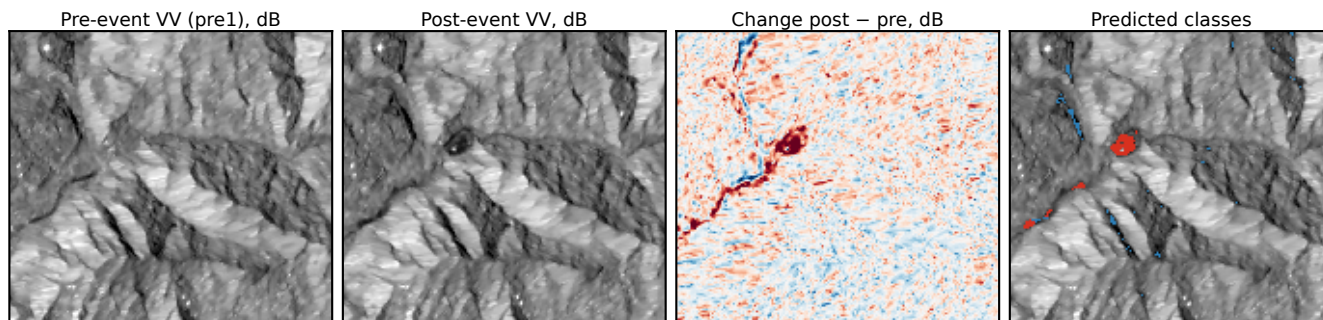


Figure 2. Trishuli sub-AOI: pre-event (pre1) and post-event VV backscatter, change post – mean(pre1, pre2) (red = darker after), and the predicted classes on the post image (red flood, blue permanent water). Model prediction, not validated.

6 Infrastructure impact and connectivity

Indicator	Count	Context
Road ways potentially affected	3	of 118; 189 m inside predicted flood
Bridges intersecting predicted flood	2	of 10 mapped
Buildings potentially affected	36	of 4,877 mapped
Settlements potentially cut off	2	of 3 analysed: Betrawati and Bhainse

- Road way: ≥ 10 m of its length inside the predicted flood (prototype rule, not calibrated).
- Bridge: intersects the predicted flood. Overlap is not evidence of damage; bridges cross water by design.
- Building: > 0.01 m² of footprint inside the predicted flood.
- Potentially cut off: disconnected in the simplified road graph after removing flood-affected edges. This is a graph-based indicator, not confirmed real-world isolation.

Settlement	Status (simplified road graph)	Reachable road before → after
Betrawati	Potentially cut off (simplified graph indicator)	31.13 → 0.70 km
Bhainse	Potentially cut off (simplified graph indicator)	31.13 → 0.39 km
Naubisephat	Already separate from main road network (before flood)	1.45 → 1.45 km

- Graph: vehicle roads in the AOI (3 edges blocked by predicted flood; main network 31.1 → 28.4 km). Footpaths are excluded, and the graph is limited to the AOI.
- Betrawati and Bhainse lose the highway link to the main network inside the AOI. The Bhainse result is especially uncertain: it sits on the AOI edge and the highway continues beyond it.

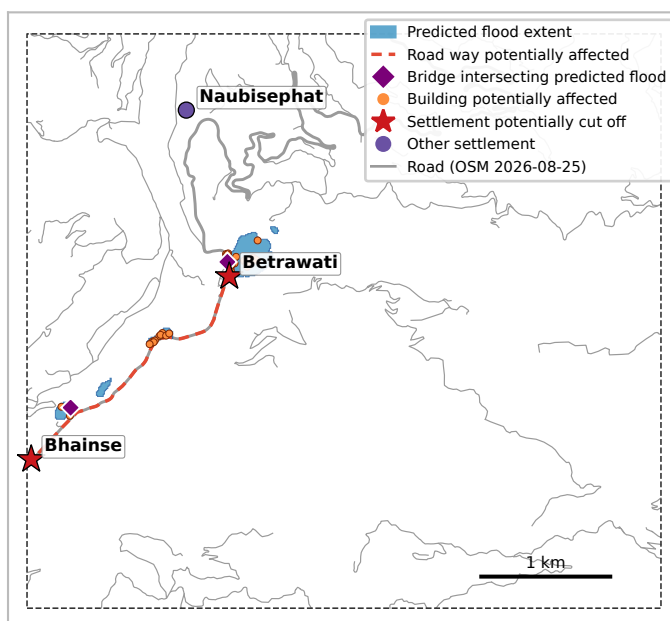


Figure 3. Infrastructure and connectivity indicators from pre-event OSM (2026-08-25) and the predicted flood. Indicators only; not confirmed damage or isolation.

7 Dashboard

- Static Next.js + MapLibre dashboard (no backend or database): interactive map with predicted flood, potentially affected road sections, bridges intersecting predicted flood, potentially affected buildings and settlements; popups, KPI cards, a settlement panel, methodology and limitations, and a “Zoom to impacts” control.
- It reads a committed, deterministic demo snapshot (~100 KB) built from the pipeline outputs. Live OpenStreetMap tiles serve as basemap for orientation only; all analytical layers come from the snapshot.

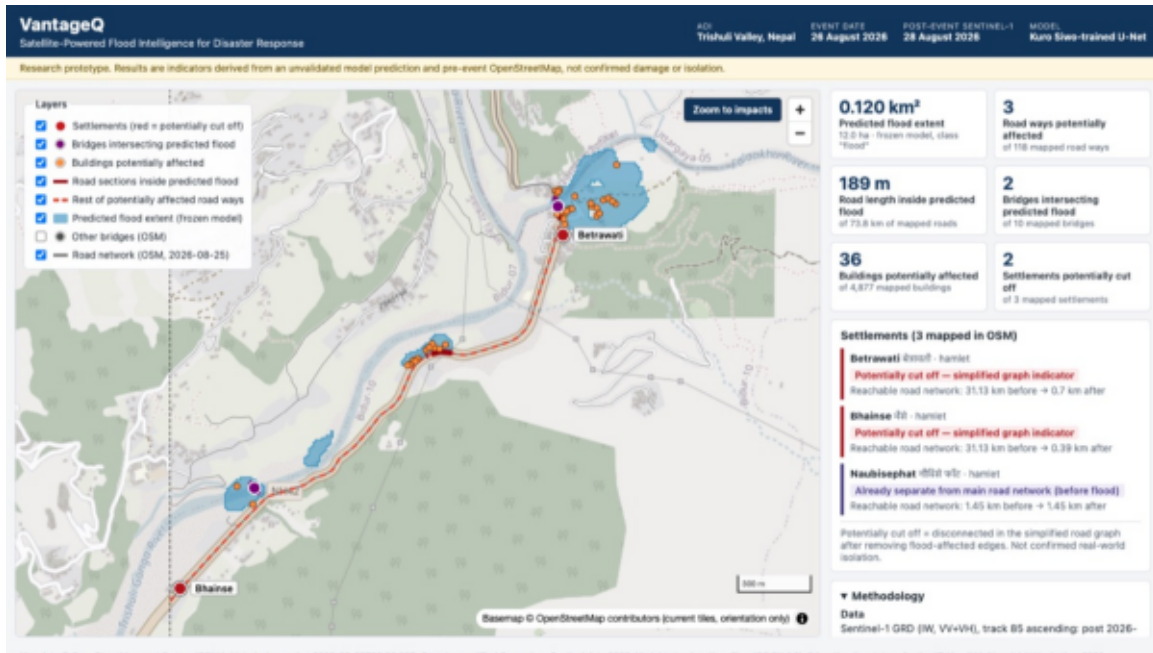


Figure 4. Dashboard after “Zoom to impacts”: impact layers on the map, KPI cards and the settlement panel with the cut-off definition.

8 Results and discussion

- On held-out Kuro Siwo events the model finds flood reliably when it predicts it (precision 0.852) but misses much of it (recall 0.324); performance varies strongly by event (flood IoU 0.066–0.691).
- Terrain-driven false positives appeared on validation (dark sloped terrain), and flood/permanent-water confusion is the largest error mode on test.
- At Trishuli, the predicted flood coincides with strong post-event darkening: pixels > 6 dB darker than the pre-event mean cover 0.55 % of the AOI (vs 0.04 % between the two pre-event dates), and 64.5 % of them are predicted flood. But only 16.4 % of pixels > 3 dB darker are predicted flood, and much of a dark trail along the valley is missed.
- The Trishuli output is therefore an unvalidated prototype indicator. VantageQ does not claim to have solved Himalayan flood mapping.

9 Limitations and future work

Limitations

- Trishuli flood map not validated against ground truth.
- Terrain effects (layover, shadow, dark slopes) can cause SAR false positives; no slope/shadow mask yet.
- Limited mountainous diversity in the training data (no Himalayan events in Kuro Siwo).
- Simplified connectivity: vehicle roads only, footpaths excluded, graph limited to the AOI.
- Bridge intersection with predicted flood is not bridge damage or impassability.
- Dashboard basemap depends on live OSM tiles (internet required).

Future work

- More labelled mountainous flood data for training and testing.
- DEM- and shadow-aware filtering of predictions.

- Broader OSM connectivity (footpaths, larger graph extent).
- Optional downstream flood-path tracing from the Copernicus DEM (planned bonus).
- Independent validation of the Trishuli result.

10 Conclusion

VantageQ is a working end-to-end prototype that turns Sentinel-1 radar imagery into a predicted flood extent and connects it to infrastructure and road-connectivity indicators, delivered through a dashboard and a one-page situation report generated from the same structured outputs. Its model results on held-out Kuro Siwo events are modest and its Trishuli output is unvalidated; the value lies in a transparent, reproducible chain from satellite data to responder-oriented indicators, with every number traceable and every limitation stated.

Developed with an AI coding assistant (Claude Code, Claude Opus 5.5). The authors directed the work and are responsible for validating all code, results and claims.